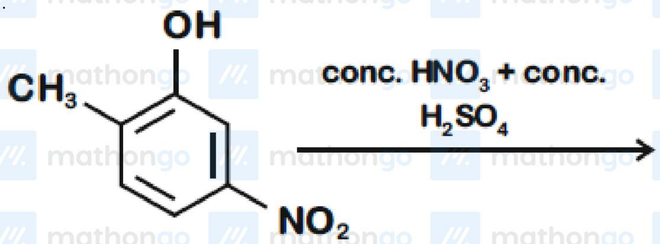
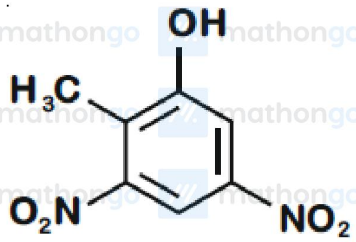
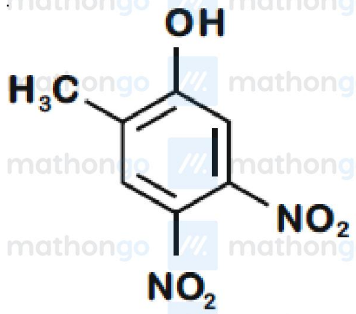
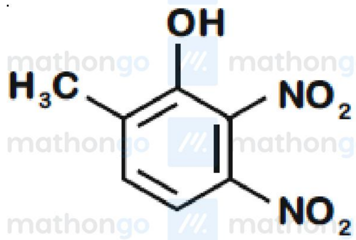
Q1 JEE Main 2020 - 2 September (Evening)

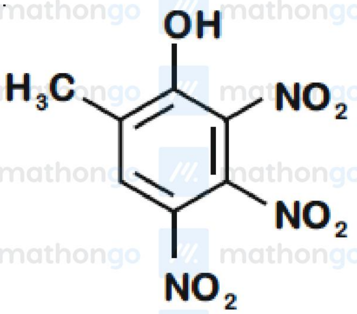
The major product of the following reaction is:



(A) 

(B) 

(C) 

(D) 

Q2 JEE Main 2020 - 3 September (Evening)

Three isomers A, B and C $(mol.$ formula $C_8H_{11}N)$ give the following results

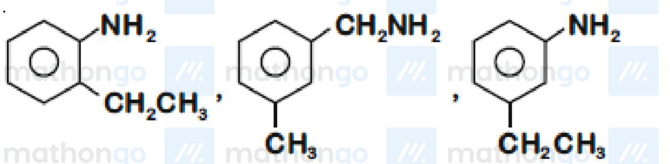
A and $\mathrm{C} \xrightarrow{\text{Diazotization}}$

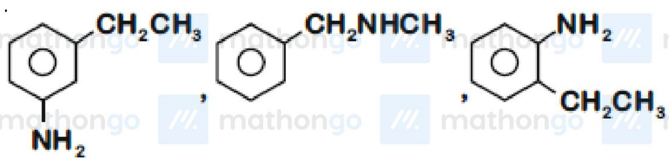
$$P + Q \xrightarrow[\text{(ii) Oxidation } (KMnO_4 + H^+)]{\text{(i) Hydrolysis}} \begin{matrix} \text{R (product of A)} \\ + \\ \text{S (product of C)} \end{matrix}$$

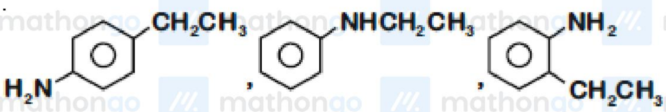
R has lower boiling point than S

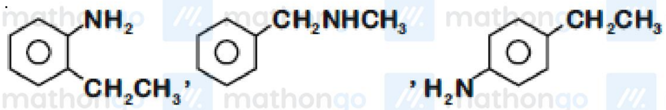
$\mathrm{B} \xrightarrow{C_6H_5SO_2Cl}$ alkali-insoluble product

$\mathrm{A, B}$ and C, respectively are

(A) 

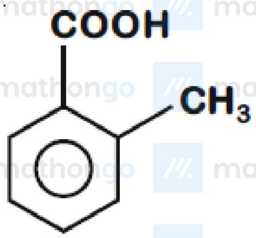
(B) 

(C) 

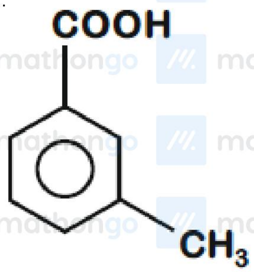
(D) 

Q3 JEE Main 2020 - 4 September (Morning)

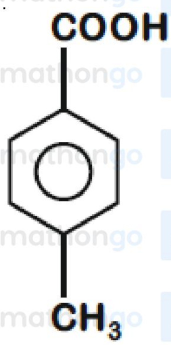
[P] on treatment with $Br_2/FeBr_3$ in CCl_4 produced a single isomer $C_8H_7O_2Br$ while heating [P] with sodalime gave toluene. The compound [P] is



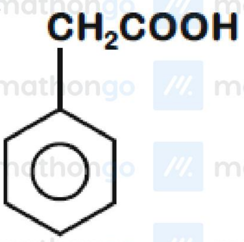
(A)



(B)



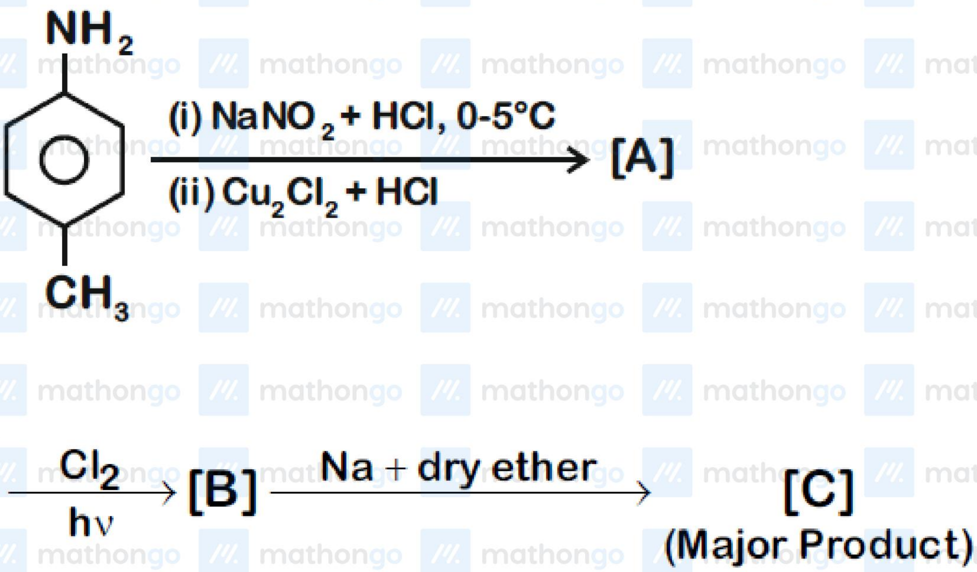
(C)

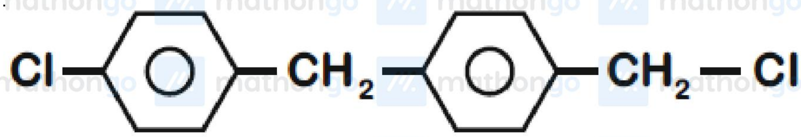


(D)

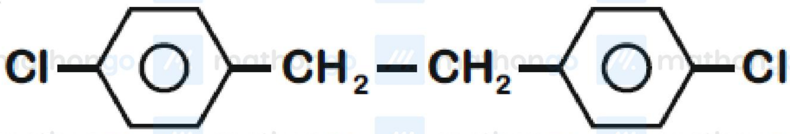
Q4 JEE Main 2020 - 4 September (Evening)

In the following reaction sequence, [C] is

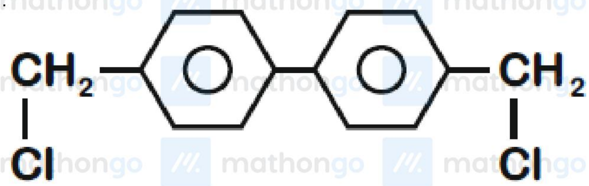




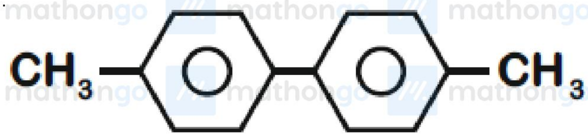
(A)



(B)



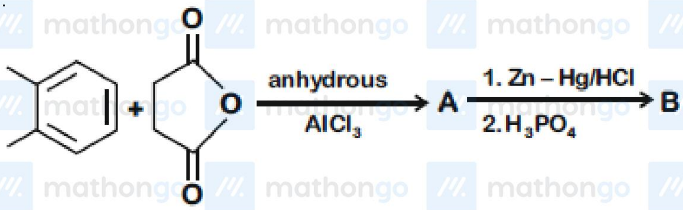
(C)

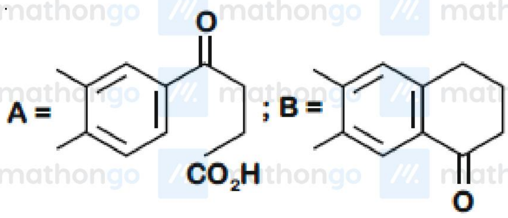


(D)

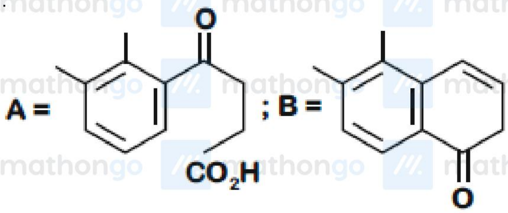
Q5 JEE Main 2020 - 5 September (Morning)

In the following reaction sequence the major products A and B are:

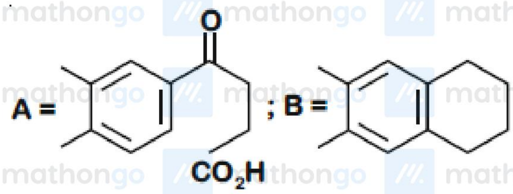




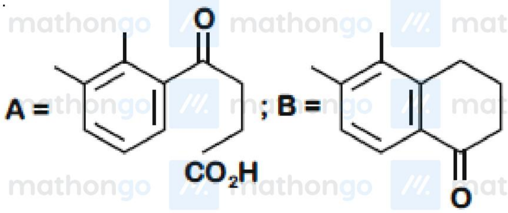
(A)



(B)



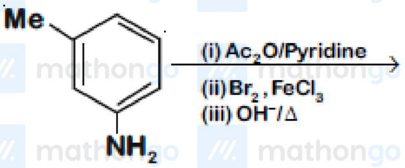
(C)

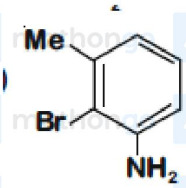


(D)

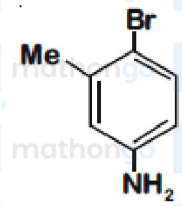
Q6 JEE Main 2020 - 5 September (Evening)

The final major product of the following reaction is

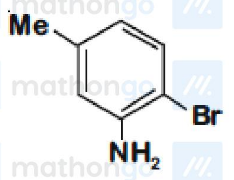


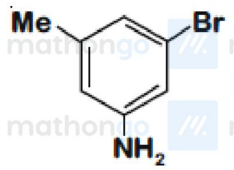


(A)



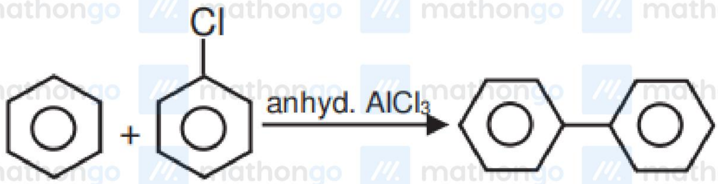
(B)

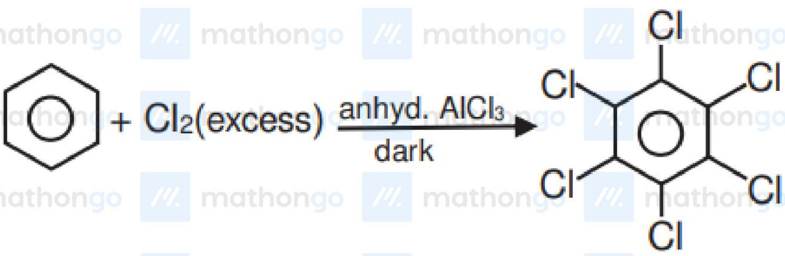
(C) 

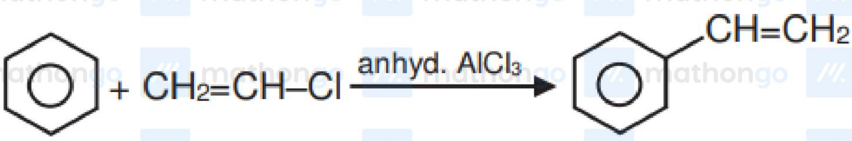
(D) 

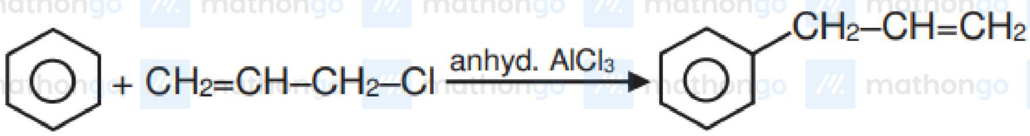
Q7 JEE Main 2020 - 7 January (Evening)

Which of the following reactions are possible ?

(a) 

(b) 

(c) 

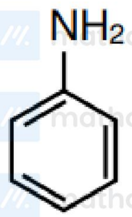
(d) 

(A) $A,\ B,\ C$
(B) $B,\ D$
(C) $A,\ C,\ D$
(D) $A,\ C$

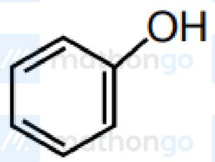
Q8 JEE Main 2020 - 9 January (Morning)

Which of these will produce the highest yield in Friedel Craft reaction ?

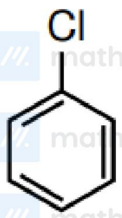
(A)



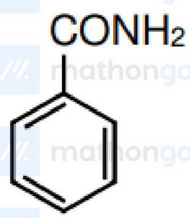
(B)



(C)

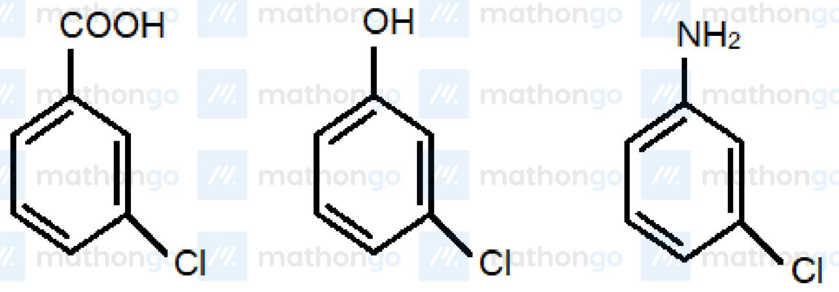


(D)



Q9 JEE Main 2020 - 7 January (Morning)

A solution of m-chloroaniline, m-chlorophenol and m-chlorobenzoic acid in ethyl acetate was extracted initially with a saturated solution of $NaNCO_3$ to give fraction A. The left over organic phase was extracted
with dilute $NaOH$ solution to give fraction B. The final organic layer was labelled as fraction C. Fraction
A, B and C, contain respectively :



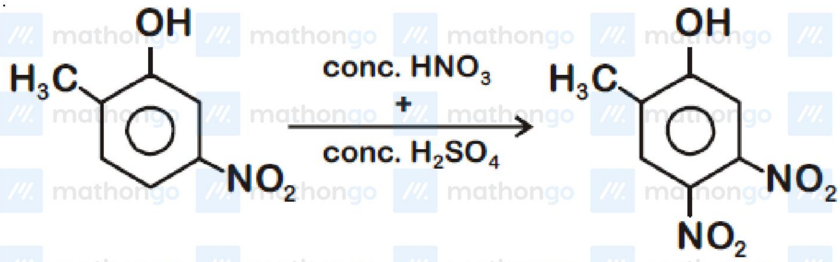
(x) (y) (z)

(A) m-chlorobenzoic acid, m-chlorophenol and m-chloroaniline
(B) m-chlorophenol, m-chlorobenzoic acid and m-chloroaniline
(C) m-chloroaniline, m-chlorobenzoic acid and m-chlorophenol
(D) m-chlorobenzoic acid, m-chloroaniline and m-chlorophenol

Answer Key

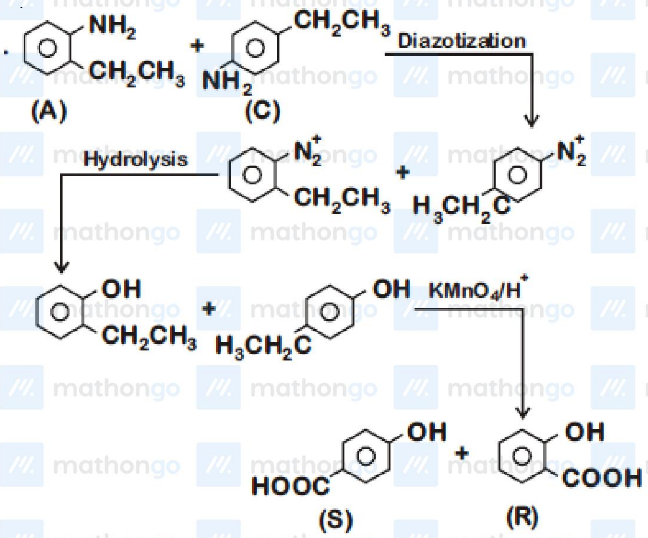
Q1 (B) **Q2** (D) **Q3** (C) **Q4** (B)
Q5 (A) **Q6** (B) **Q7** (B) **Q8** (C)
Q9 (A)

Q1

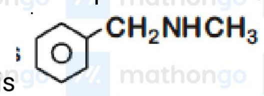


Position of electrophilic attack is directed by the electron donating group present in ring

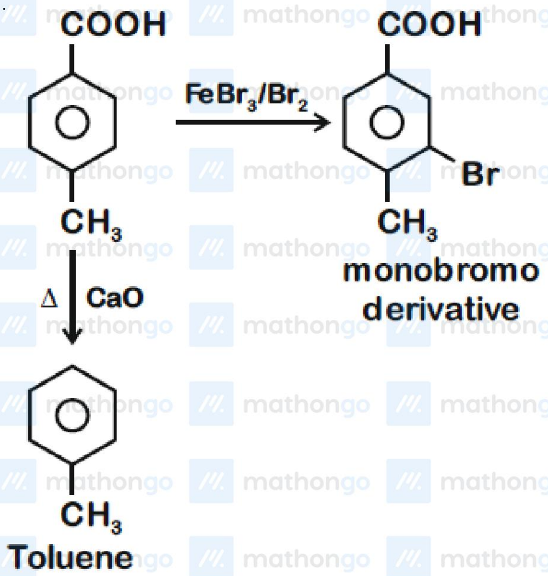
Q2



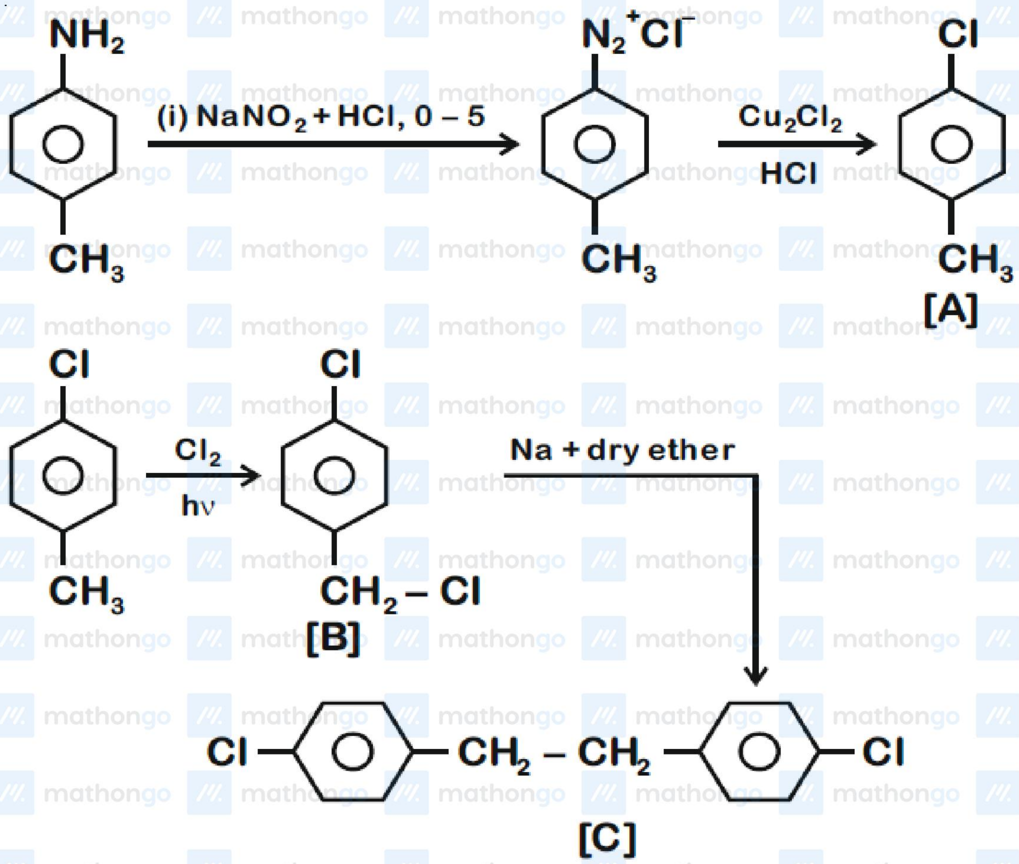
(B) gives insoluble product with $C_6H_5SO_2Cl$.

Hence (B) is 

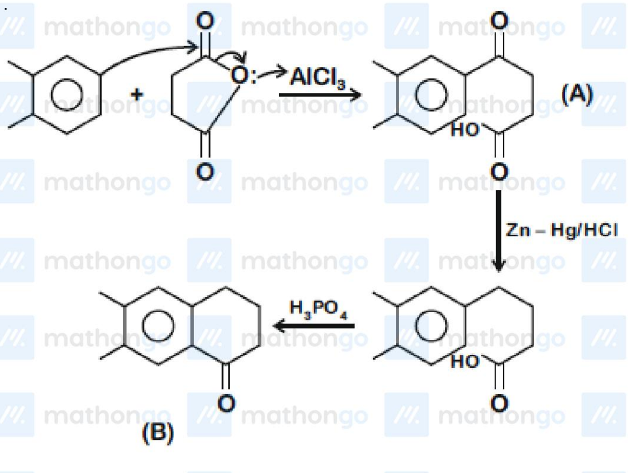
Q3


COOH
FeBr3/Br2
COOH
Br
CH3
CH3
monobromo
derivative
Δ
CaO
CH3
Toluene

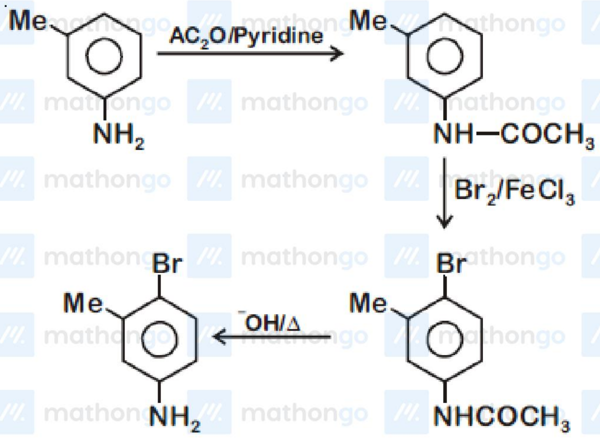
Q4


NH2
(i) NaNO2 + HCl, 0 – 5
N2+Cl–
Cu2Cl2
HCl
Cl
CH3
[A]
Cl2
hν
CH2 – Cl
[B]
Na + dry ether
Cl
CH2 – CH2
[C]

Q5


O
O:
AlCl3
(A)
HO
Zn – Hg/HCl
H3PO4
(B)

Q6



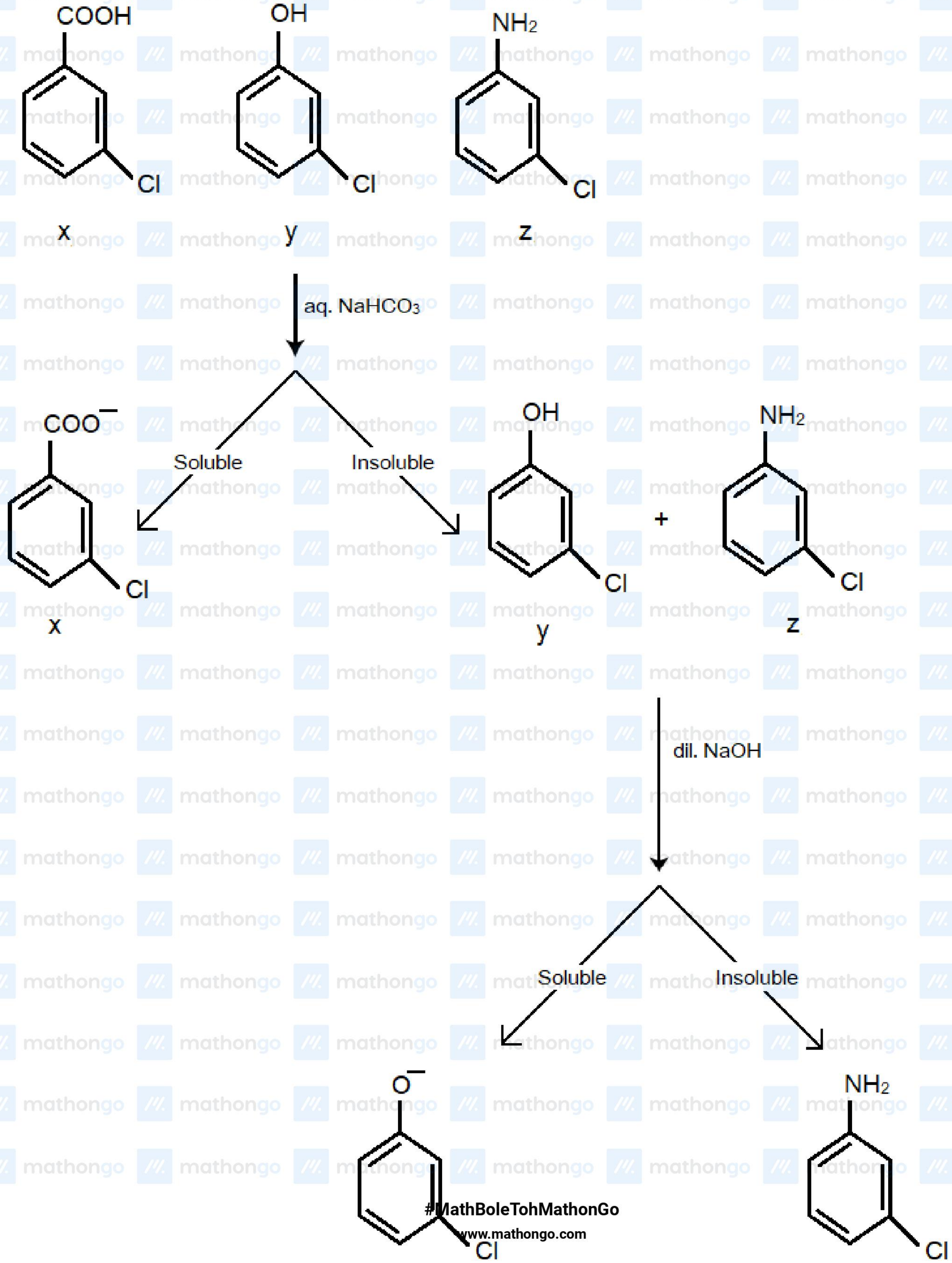
Q7

Vinyl halides and aryl halides do not give Friedel craft's reaction.

Q8

Aniline and phenol form complex with lewis acid so most reactive among the given compounds for Friedel Craft reaction is chlrobenzene.

Q9


COOH
Cl
x
OH
Cl
y
NH2
Cl
z
aq. NaHCO3
Soluble
Insoluble
COO−
Cl
x
OH
Cl
y
+
NH2
Cl
z
dil. NaOH
Soluble
Insoluble
O−
Cl
NH2
Cl